

Adaptive Momentum Memory for Temporal Planning Consistency in End-to-End Autonomous Driving (MemAd)

Project Proposal - Davron M & Alex R



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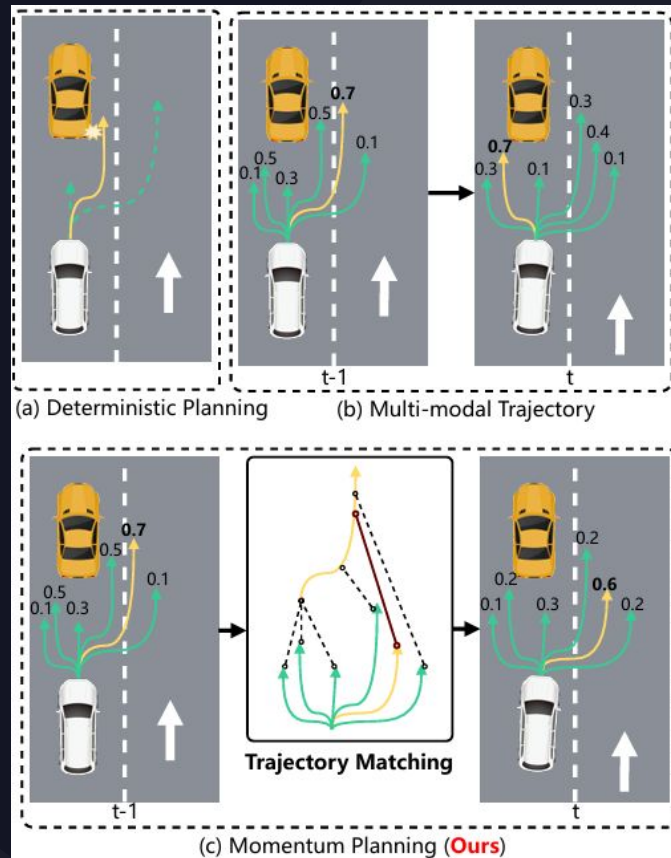
01

Background



MomAD

- **End-to-End (E2E) Framework:**
Raw sensor data directly into vehicle motion plans
- **The Problem: Causal Confusion:**
Past velocity—rather than actual environmental cues
- **Momentum Integration:**
Temporal consistency in decision-making.



Cont. MomAD

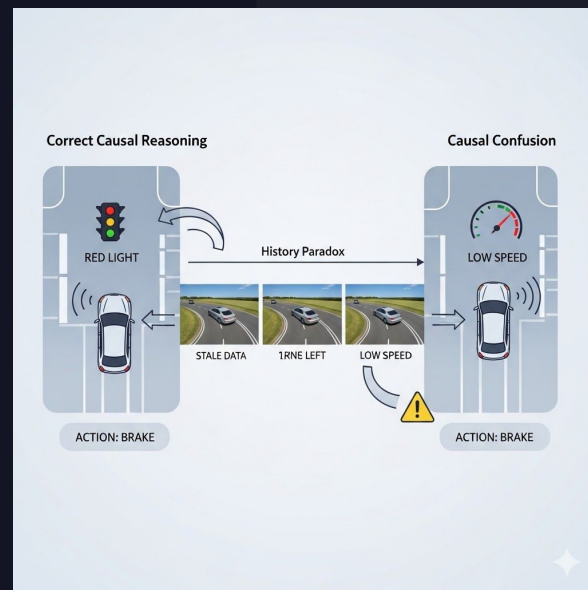
$$\text{TPC} = \frac{1}{N_T} \sqrt{\sum_{i=1}^N \left((T_t^{\text{Pred}} - T_{t-1}^{\text{Pred}})^2 \cdot T_{\text{GT}}^{\text{Mask}} \right)}, \quad (6)$$

where N_T is the total number of GT trajectories in the validation set, and $T_{\text{GT}}^{\text{Mask}}$ is the mask for trajectories exceeding the overlapped time period of two trajectories. Our TPC metric evaluates whether autonomous vehicles adhere to predicted trajectories, ensuring continuity across frames. Notably, the TPC metric provides a statistical perspective on the dataset-wide evaluation rather than at the individual sample level.

- **Topological Trajectory Matching (TTM)**
Path is aligned with the vehicle's historical trajectory
- **Momentum Planning Interactor (MPI)**
Fuse current queries with past historical data
- **Key Metric: TPC**
Trajectory Prediction Consistency

Current challenges

- **The "History Paradox":** Ablation studies show that increasing history length (t) does not linearly improve performance; specifically, $t=2$ outperformed $t=3$, indicating that more history can degrade planning quality.
- **Causal Confusion (The Copycat Problem):** Models frequently latch onto "spurious correlations," such as current velocity or past trajectory, instead of reacting to immediate visual cues like traffic lights.



02

Improvement



Adaptive History Length for Momentum Planning

- Limitation
 - MPI uses a fixed amount of historical information, even though the best history length may vary by scene.
- Hypothesis
 - The optimal planning history should depend on scene context and planning uncertainty.
- Improvement
 - Adaptive history selector for MPI that chooses whether to use: no history, 1 previous frame, 2 previous frames

Table 7. Impact of history frames in MomAD on the Turning-nuScenes validation set. t denotes the frame number, where 1 indicates the history is empty (represented by 0), 2 signifies that the historical result corresponds to the previous 1 frame, and 3 indicates that the historical result pertains to the previous 2 frames. We follow the VAD [19] evaluation metric.

ED	MP	NS	t	L2(m) ↓			Col. Rate(%) ↓			TPC(m) ↓		
				2s	3s	Avg.	2s	3s	Avg.	2s	3s	Avg.
				0.77	1.46	0.86	0.17	0.98	0.40	0.70	1.33	0.79
	✓		2	0.71	1.27	0.77	0.14	0.83	0.34	0.55	1.07	0.65
✓		0.1		0.77	1.45	0.86	0.18	0.97	0.39	0.69	1.33	0.79
✓	✓	0.1	1	0.78	1.48	0.88	0.18	0.98	0.39	0.70	1.35	0.81
✓	✓	0.1	2	0.70	1.24	0.76	0.13	0.79	0.32	0.54	1.05	0.63
✓	✓	0.1	3	0.72	1.27	0.78	0.14	0.84	0.35	0.56	1.09	0.66

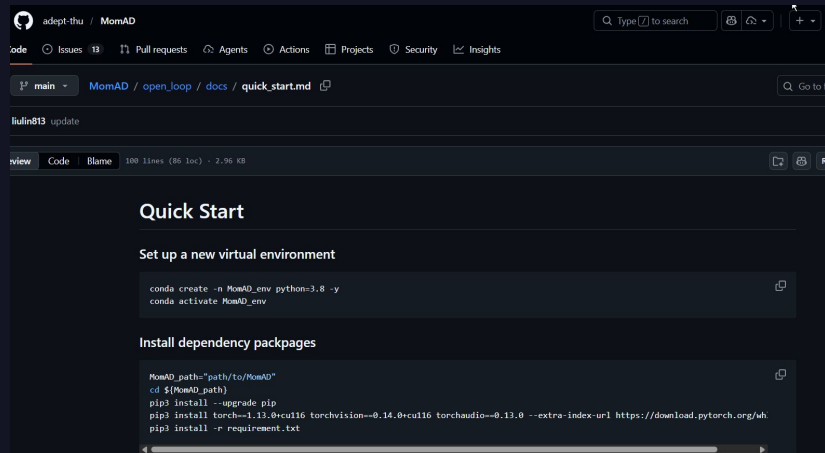
03

Methods



Overall Plan

1. Reproduce official MomAD baseline
2. Keep TTM, denoising, and loss definitions unchanged
3. Modify only the MPI temporal-history pathway
4. Verify performance of history usage being adaptive per scene instead of fixed



Current MPI Pipeline

- Previous planning score
- Previous planning query
- Current selected planning query

Long Horizon Query Mixer always tries to enrich the current query with the available past planning state

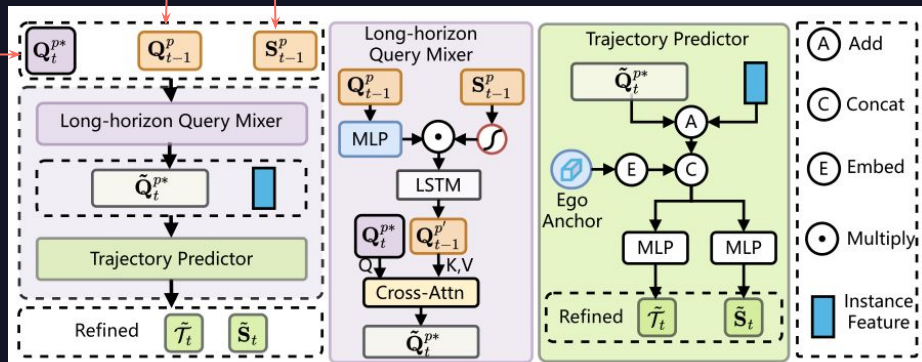
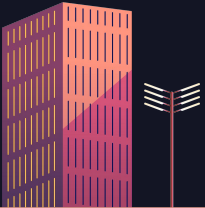
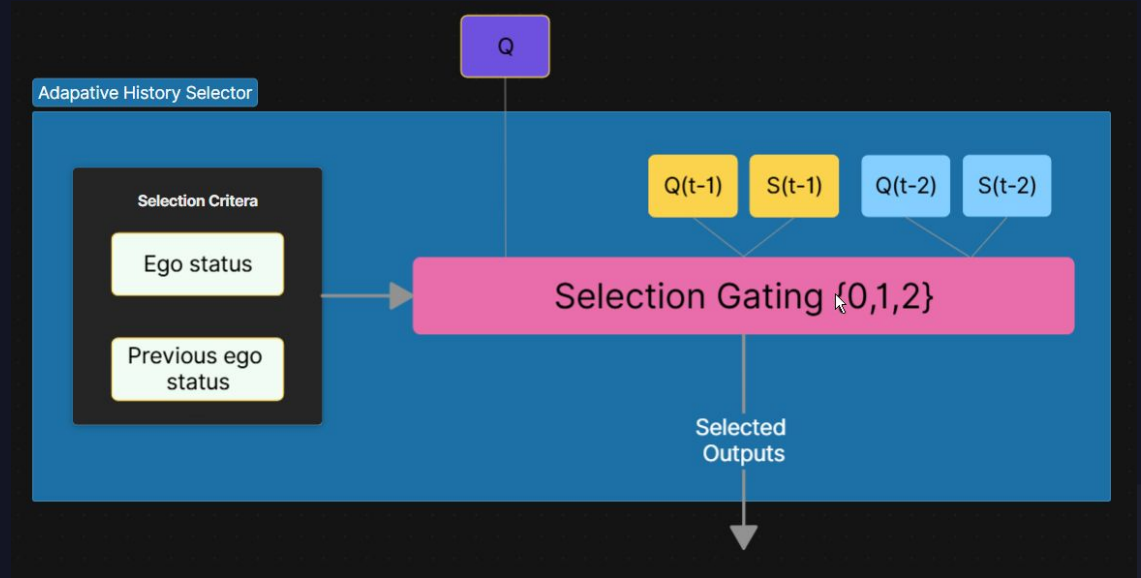


Figure 3. **The illustration of Momentum Planning Interactor (MPI).** MPI cross-attends a selected planning query with historical queries to expand static and dynamic perception files, resulting in an enriched query that improves long-horizon trajectory generation and reduces collision risks.

Adaptive History Selector

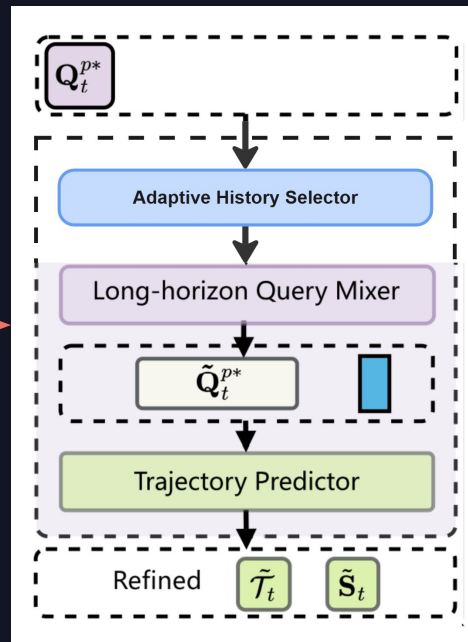
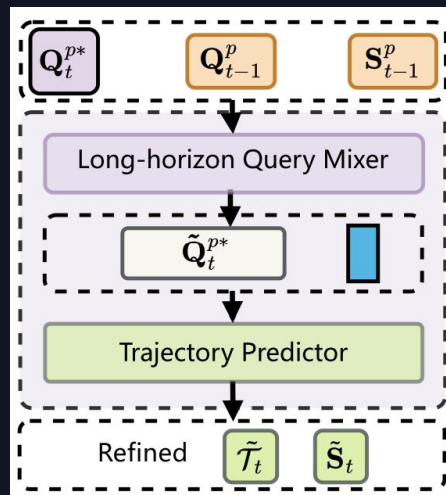
Ego (car) status data

- Steering angle
- Rotation rate
- Velocity
- Acceleration



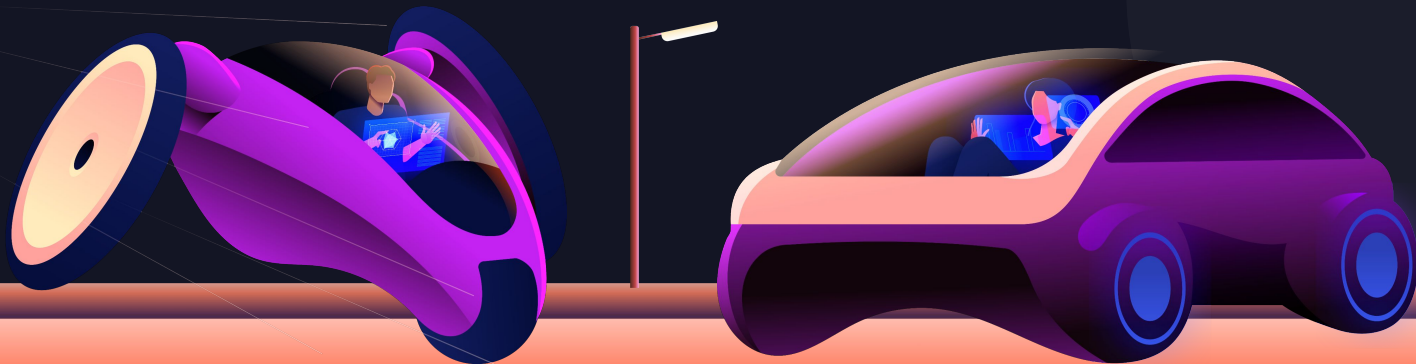
Planned MPI Pipeline

- New module: Adaptive History Selector
 - Insert a lightweight selector before the Long-horizon Query Mixer
- Selector predicts history depth
 - 0, 1, or 2



04

Verification



Verification process

Comparison

31%

Q3
31.0%

Implementation

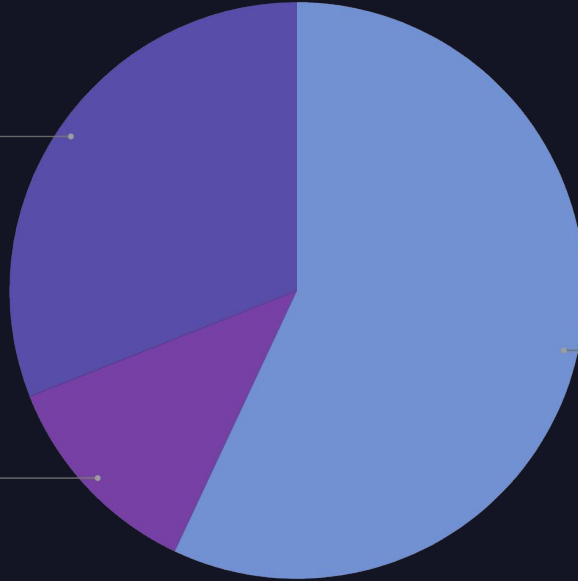
12%

Q2
12.0%

Simulation

57%

Q1
57.0%



Phase I: Closed-Loop Simulation

Primary Benchmark

Conduct extensive testing on the **Bench2Drive** and **nuPlan** closed-loop simulators to evaluate reactive driving behavior.

Failure Analysis

Document cases where the **Adaptive History Selector** successfully "forgets" stale data to avoid the **Momentum Trap**

Scenario Focus

Prioritize the **"Turning-nuScenes"** subset, where temporal inconsistency and the "History Paradox" are most prevalent.

Core Metrics

Measure success via **L2 Distance** (path accuracy), **Collision Rate**, and specifically **TPC (Trajectory Prediction Consistency)**

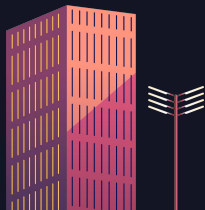
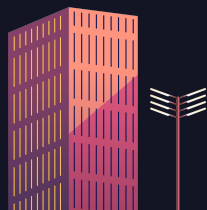


Table 3. Long trajectory planning results on the nuScenes and Turning-nuScenes validation sets. We train models for 10 epochs for 6s-horizon prediction. T-nuScenes indicates the challenging Turning-nuScenes. We follow the VAD [21] evaluation metric.

Split	Method	L2 (m) ↓			Col. Rate (%) ↓			TPC (m) ↓		
		4s	5s	6s	4s	5s	6s	4s	5s	6s
nuScenes	SparseDrive [30]	1.75	2.32	2.95	0.87	1.54	2.33	1.33	1.66	1.99
	MomAD	1.67	1.98	2.45	0.83	1.43	2.13	1.19	1.45	1.61
		-0.09	-0.34	-0.50	-0.04	-0.11	-0.20	-0.14	-0.21	-0.38
T-nuScenes	SparseDrive [30]	2.07	2.71	3.36	0.91	1.71	2.57	1.54	2.31	2.90
	MomAD	1.80	2.07	2.51	0.85	1.57	2.31	1.37	1.58	1.93
		-0.27	-0.64	-0.85	-0.06	-0.14	-0.26	-0.17	-0.73	-0.97



Phase II: SOTA Comparison & Ablation

Baseline Benchmarking

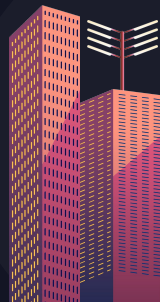
Compare "Adaptive MomAd" against the original **MomAd** baseline using fixed history lengths of $t=1$, $t=2$, and $t=3$.

Statistical Significance

Perform an **Ablation Study** to prove that the Gating Network's decisions are mathematically superior to static windows

SOTA Alignment

Contrast results with other state-of-the-art models mentioned in the survey, such as **TransFuser** or **UniAD**



Phase III: Real-World Go-Kart Implementation

Edge Deployment

Port the trained model to the **Raspberry Pi 5** using **ROS2 Jazzy Jalisco** for real-time inference.

Sim-to-Real Transfer

Validate technical feasibility by measuring **FPS (Frames Per Second)**, **Inference Latency**, and power consumption

Sim-to-Real Transfer

Conduct low-speed track tests to verify that the adaptive memory helps the go-kart navigate sharp turns without "jittering" due to stale visual data



Acknowledgements

<https://doi.org/10.48550/arXiv.2503.03125>



Thanks!

Do you have any questions?

